GEOMETRY-AWARE QUANTUM CLOUD CHALLENGE

A Verified Bell Pair on Three Processor Models

Qiskit Fall Fest 2026 Hackathon | Basic Track | Builder-level study

EXECUTIVE SUMMARY

The project compares ideal, noisy, and ZZ-protected Bell-state circuits across 2016-inspired, contemporary, and hyperbolic-inspired quantum processor models. In the saved core run, the ZZ check clearly improves fidelity on the contemporary model, but produces essentially no gain on the other two models relative to the uncertainty. The check adds three two-qubit gates, one SWAP, extra depth, and discards some shots. Common-noise and ancilla-placement experiments indicate that connectivity-aware, calibration-aware compilation is central to getting value from protection.

Prepared from the supplied challenge brief, notebook, saved CSV tables, settings, and figures.

1. Problem Statement and Objectives

The challenge requires a verified Bell correlation between two fixed output ports on each of three prescribed processor models. Each model is run under ideal conditions, noisy conditions, and noisy conditions with one justified protection technique. Required metrics include fidelity and uncertainty, circuit depth, two-qubit gate count, SWAP overhead, physical-qubit usage, acceptance fraction, and discarded shots.

The central question is: when does an error-detection circuit improve Bell-state fidelity enough to justify its additional gates, routing, ancilla use, and discarded measurements? A second question is whether observed differences are caused by coupling geometry or by changes in gate sets and noise assumptions.

The notebook completes the seven challenge tasks and includes Builder-level extensions: a candidate-ancilla placement study, a ZZ+XX double-check experiment, a common-noise graph comparison, and a two-qubit-error sensitivity sweep.

2. Processor Models and Reproducibility

Processor model	Fixed ports	Role	Noise treatment
2016-inspired	0 and 4	Historical-style reference	FakeBurlingtonV2-based noise setup
Contemporary	126 and 129	Current compilation baseline	FakeFez calibration/target data
Hyperbolic-inspired	0 and 3	Hypothetical alternative geometry	Synthetic circuit-level noise

The 2016-inspired model uses FakeBurlingtonV2; it is not calibration data recorded in 2016. The contemporary model uses FakeFez. The hyperbolic-inspired model represents a finite hypothetical architecture, not a deployed device. Native results combine graph, gate-set, and noise differences; they should not be interpreted as a geometry-only comparison.

Reproducibility settings: Python 3.11.15; Qiskit 2.4.2; Qiskit Aer 0.17.2; qiskit-ibm-runtime 0.50.0; 10,000 shots per setting; transpiler seed 1; simulator seed 1; optimization level 2. Each case combines X, Y, and Z measurement settings, for 30,000 shots before post-selection.

3. Methodology

3.1 Bell-state preparation and routing

The logical circuit applies a Hadamard gate to the first port qubit, followed by a CNOT to the second port qubit. The target is $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. Because the two ports are not direct neighbours, Qiskit maps the logical circuit onto each coupling graph and decomposes it into that model's supported gate set. The unprotected circuit uses 7 two-qubit gates and 2 SWAPs on all three models. Its depth is 7 for the 2016-inspired model and 14 for the contemporary and hyperbolic-inspired models.

3.2 Bell-state verification

Z-basis agreement alone is insufficient because a classical mixture of $|00\rangle$ and $|11\rangle$ can show correlated outcomes without being the target entangled state. We therefore estimate XX, YY, and ZZ. X is measured by applying H to both ports before Z-basis measurement; Y is measured by applying S† and then H to both ports; Z is measured directly.

$$F(\Phi^+) = [1 + \langle XX\rangle - \langle YY\rangle + \langle ZZ\rangle] / 4$$

For each setting, the correlator is the even-parity fraction minus the odd-parity fraction. The standard error is $\sigma_{PP} = \sqrt{[(1 - \langle PP\rangle^2)/N]}$, where N is the accepted sample count for that setting. The fidelity uncertainty is $\sigma_F = \frac{1}{4}\sqrt{(\sigma_{XX}^2 + \sigma_{YY}^2 + \sigma_{ZZ}^2)}$. The benchmark treats the state as verified if $F - 2\sigma_F > 0.5$. All nine core cases pass this saved-output criterion.

3.3 ZZ parity check and post-selection

The ZZ check adds one ancilla prepared in $|0\rangle$. CNOT gates connect each port to the ancilla, which is then measured. An ancilla outcome of 0 indicates the expected ZZ parity (+1), so the analysis keeps only those shots. The check can detect certain X or Y errors that flip ZZ parity, but not every error, including a single-port Z error.

Protected fidelity is conditional on passing the check. It must be reported together with acceptance and discarded-shot count. Post-selection is a trade-off, not a free fidelity gain.

3.4 Builder-level experiments

- Ancilla placement: candidate locations are ranked using compiled two-qubit gate count and calibration-based success estimates as a tie-breaker; selection is not tuned on observed fidelity.
- ZZ+XX double check: a second ancilla and an XX parity check are added to target another class of errors.
- Common-noise and sensitivity studies: all three graphs are compiled with a shared gate set and synthetic noise, followed by a sweep over two-qubit error rates. These are supplementary studies, not replacements for the required native-noise benchmark.

4. Core Results: the Nine Required Cases

Table 1 reports the saved core comparison. Fidelity is shown as F ± σ_F. Acceptance is the share of the 30,000 attempted shots retained after post-selection.

Case	Model	Scenario	F ± σF	F−2σ	Depth	2Q	SWAPs	Qubits	Accepted	Discarded
1	2016	Ideal baseline	1.0000 ± 0.0000	1.0000	7	7	2	4	100.00%	0
2	2016	Noisy baseline	0.8387 ± 0.0027	0.8333	7	7	2	4	100.00%	0
3	2016	Noisy + ZZ check	0.8383 ± 0.0029	0.8325	13	10	3	5	88.62%	3,414
4	Contemp.	Ideal baseline	1.0000 ± 0.0000	1.0000	14	7	2	4	100.00%	0
5	Contemp.	Noisy baseline	0.9460 ± 0.0016	0.9428	14	7	2	4	100.00%	0
6	Contemp.	Noisy + ZZ check	0.9704 ± 0.0012	0.9680	23	10	3	5	95.94%	1,219
7	Hyperbolic	Ideal baseline	1.0000 ± 0.0000	1.0000	14	7	2	4	100.00%	0
8	Hyperbolic	Noisy baseline	0.9520 ± 0.0015	0.9490	14	7	2	4	100.00%	0
9	Hyperbolic	Noisy + ZZ check	0.9543 ± 0.0015	0.9513	29	10	3	4	96.77%	968

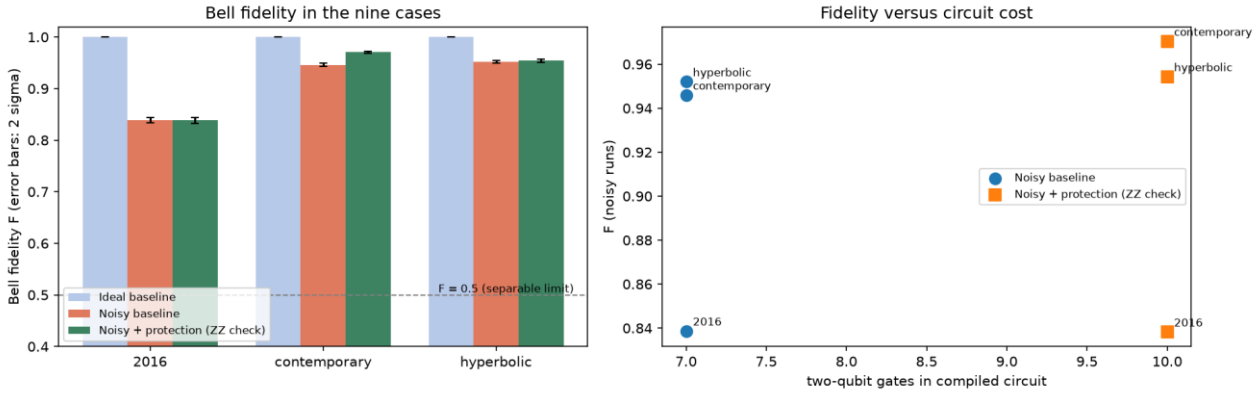


Figure 1. Fidelity in the nine core cases. The dashed horizontal line marks F = 0.5, and error bars represent two standard errors.

4.1 Main interpretation

- 2016-inspired: fidelity changes from 0.8387 to 0.8383 (ΔF ≈ −0.0004), negligible relative to uncertainty. Depth rises from 7 to 13, two-qubit gates from 7 to 10, and SWAPs from 2 to 3. The check discards 3,414 of 30,000 shots (11.38%). In this saved core run the check does not pay for itself.
- Contemporary: fidelity rises from 0.9460 to 0.9704 (ΔF ≈ +0.0244). Depth rises from 14 to 23, gates from 7 to 10, and SWAPs from 2 to 3. The check discards 1,219 shots (4.06%). This is the clearest improvement in the core comparison.
- Hyperbolic-inspired: fidelity rises from 0.9521 to 0.9543 (ΔF ≈ +0.0023). The 96.77% acceptance is high, but depth increases from 14 to 29. The observed gain is small relative to the combined uncertainty scale, so it should not be described as decisive.

The contemporary result provides the strongest evidence that the single ZZ check is useful in the tested setup. The other two outcomes show why fidelity must be interpreted alongside uncertainty, overhead, and data yield.

5. Compiled Circuit Diagrams

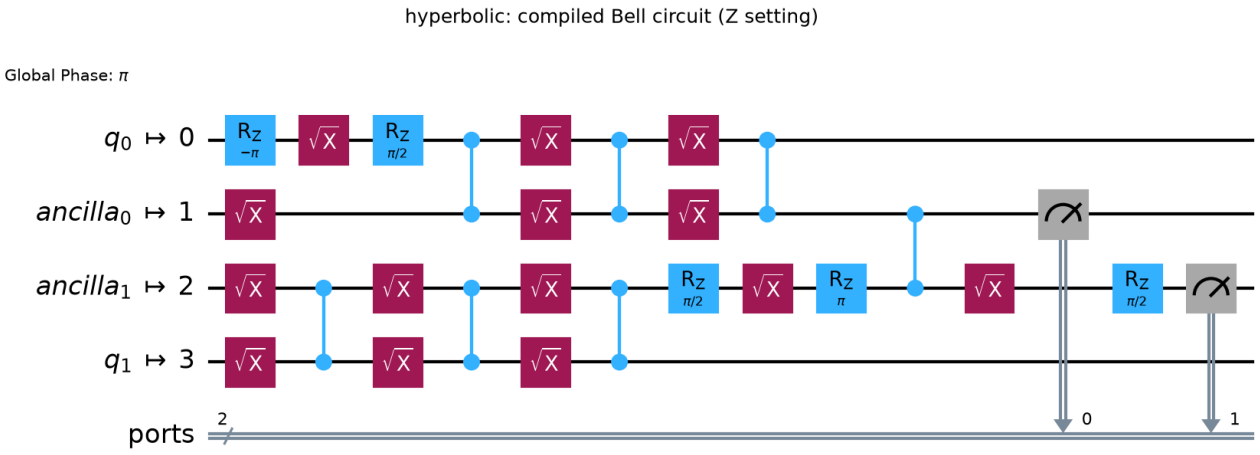


Figure 2. Hyperbolic-inspired compiled Bell circuit for the Z measurement setting, without the explicit ZZ parity check.

The simple logical Hadamard-CNOT circuit becomes longer after mapping and gate decomposition. The additional physical wires labelled as ancillas by the transpiler should not automatically be interpreted as logical error-check ancillas; routing/layout qubits may also appear in the compiled drawing.

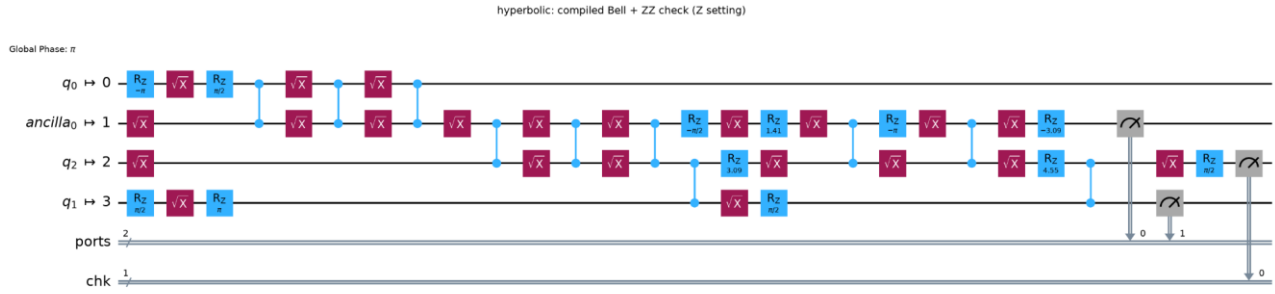


Figure 3. Hyperbolic-inspired compiled Bell circuit with the additional ZZ parity check.

The protected diagram includes the check ancilla and measurement. Rz and √X gates are part of the backend-compatible decomposition. The displayed global phase π does not affect measurement probabilities. In the saved hyperbolic compilation, the check increases two-qubit gates from 7 to 10 and depth from 14 to 29; the total unique physical-qubit count remains four because the check ancilla is on a qubit already included in the routed physical set.

6. Builder-Level Study: Ancilla Placement and Double Checks

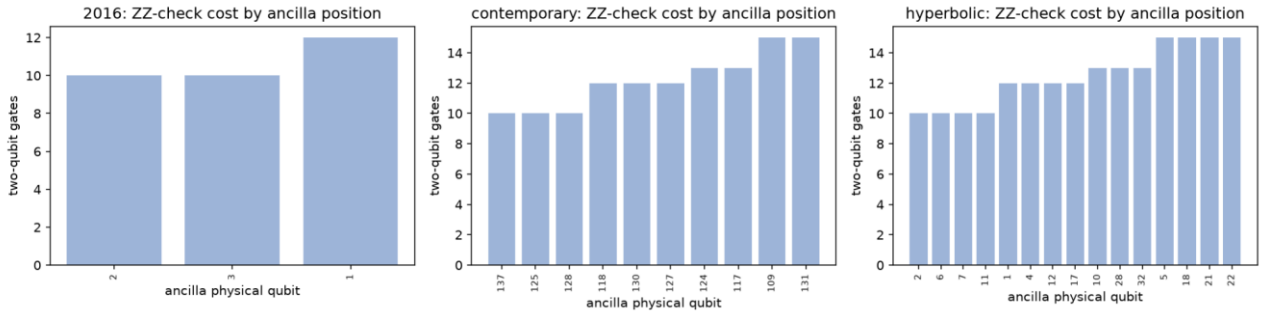


Figure 4. Compiled two-qubit gate count for candidate ancilla locations. Gate count is a cost metric, not a guarantee of fidelity.

Model	Variant	Ancilla	F	ΔF vs base	Accept	Depth	2Q	SWAP
2016-inspired	Higher-cost candidate	[1]	0.8412	+0.0025	86.78%	14	12	3
2016-inspired	Cost-selected candidate	[2]	0.8383	-0.0004	88.62%	13	10	3
2016-inspired	ZZ+XX double check	[2,3]	0.8259	-0.0128	74.70%	17	17	4

Model	Variant	Ancilla	F	ΔF vs base	Accept	Depth	2Q	SWAP
Contemporary	Higher-cost candidate	[131]	0.9558	+0.0098	95.61%	31	15	4
Contemporary	Cost-selected candidate	[137]	0.9704	+0.0244	95.94%	23	10	3
Contemporary	ZZ+XX double check	[137,125]	0.9421	-0.0039	94.42%	40	16	5
Hyperbolic-inspired	Higher-cost candidate	[22]	0.9448	-0.0072	96.44%	31	15	4
Hyperbolic-inspired	Cost-selected candidate	[2]	0.9543	+0.0023	96.77%	29	10	3
Hyperbolic-inspired	ZZ+XX double check	[2,12]	0.9655	+0.0134	93.14%	37	14	4

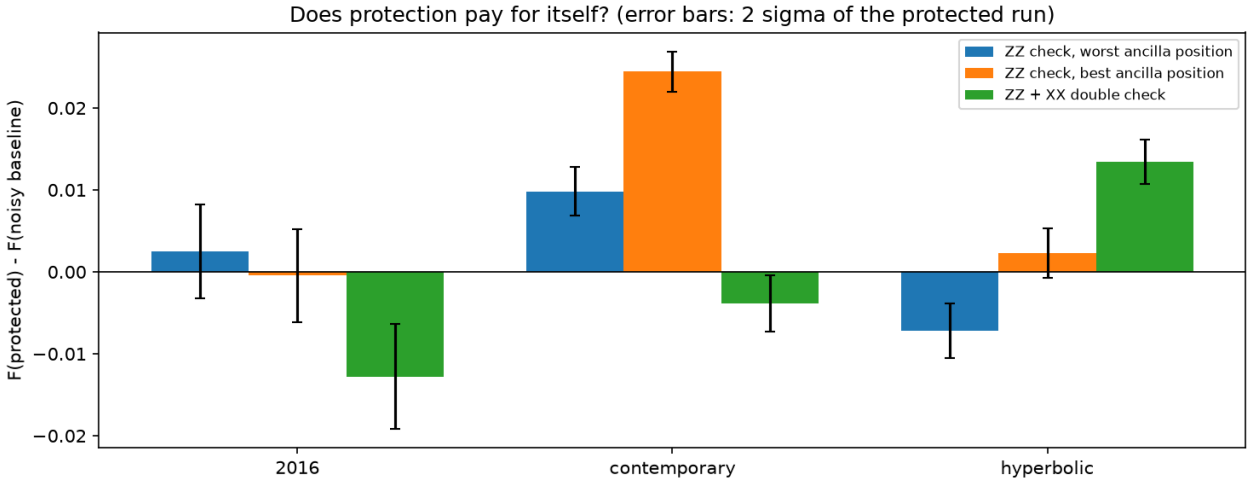


Figure 5. Fidelity change relative to the native noisy baseline for the selected placement and double-check variants.

The saved CSV demonstrates that candidate placement changes both compiled cost and fidelity. The cost-selected single-check candidates use 10 two-qubit gates, while higher-cost alternatives use 12–15. The cost-selected contemporary candidate (ancilla 137) has F = 0.9704; the higher-cost candidate (ancilla 131) has F = 0.9558. For the hyperbolic model, the cost-selected candidate (ancilla 2) has F = 0.9543, while the higher-cost candidate (ancilla 22) has F = 0.9448. For the 2016-inspired model, the higher-cost candidate F = 0.8412 is slightly above the cost-selected candidate F = 0.8383, so the minimum-cost choice is not the highest-fidelity outcome in that case.

The ZZ+XX double check is not universally better. Its saved fidelity is 0.8259 on the 2016-inspired model and 0.9421 on the contemporary model, both below the corresponding unprotected baseline. On the hyperbolic-inspired model it reaches 0.9655, but acceptance drops to 93.14% and the circuit is deeper. Additional checks can reject more unwanted outcomes, but their own gates can introduce additional error.

7. Common-Noise Comparison: Isolating Connectivity

The native runs mix graph geometry with different gate sets and noise assumptions. The notebook therefore reruns all three graphs using the same gate set and synthetic noise parameters. In this supplemental test, every unprotected graph has F = 0.9521, depth 14, 7 two-qubit gates, and 2 SWAPs.

Model	Protected F ± σ	F−2σ	Accept	Depth	2Q gates	SWAPs	Qubits
2016-inspired	0.9577 ± 0.0015	0.9548	97.00%	23	10	3	5
Contemporary	0.9672 ± 0.0013	0.9646	96.08%	28	10	3	5
Hyperbolic-inspired	0.9543 ± 0.0015	0.9513	96.77%	29	10	3	4

Under common noise, the protected fidelity is 0.9577 for the 2016-inspired graph, 0.9672 for the contemporary graph, and 0.9543 for the hyperbolic-inspired graph. The contemporary graph has the highest mean fidelity and a protected depth of 28, compared with 23 and 29 for the other two graphs. This suggests graph structure affects the protected compilation in this finite test. It does not establish a universal architecture ranking.

8. Sensitivity to Two-Qubit Error

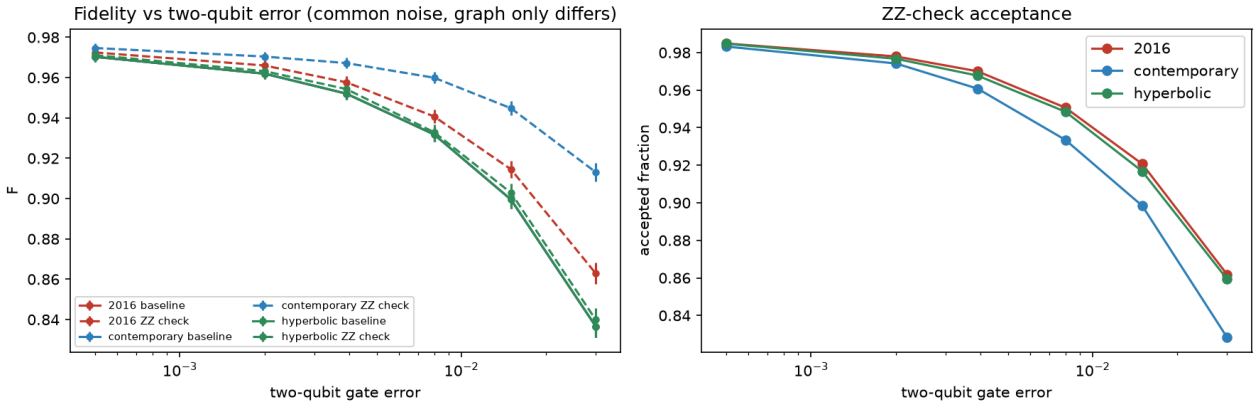


Figure 6. Fidelity (left) and ZZ-check acceptance fraction (right) across the tested two-qubit error rates; horizontal axis is logarithmic.

The sweep varies synthetic two-qubit gate error from 0.05% to 3%, while keeping the graph-only comparison and other sweep settings fixed. Fidelity decreases as gate error increases in both the baseline and protected cases. At the central 0.39% setting, the observed ZZ-check gains are approximately +0.0056 (2016-inspired), +0.0152 (contemporary), and +0.0023 (hyperbolic-inspired). At 3% error the corresponding mean gains are about +0.0263, +0.0765, and +0.0033.

Acceptance falls as two-qubit error rises. At 3% error, acceptance is approximately 86.19% on the 2016-inspired graph, 82.84% on the contemporary graph, and 85.94% on the hyperbolic-inspired graph. The contemporary model shows a strong nominal benefit through the sweep. The hyperbolic improvements are much smaller and overlap the statistical-uncertainty scale at several points, so the sweep does not establish a statistically robust gain at every error rate.

9. Discussion and Design Recommendation

The native baseline results differ substantially even though all three compiled unprotected circuits use seven two-qubit gates and two SWAPs. The most direct explanation is that these native runs use different noise assumptions and gate sets; the native comparison cannot assign the fidelity gap to geometry alone. The common-noise study is important because it controls those assumptions and shows how topology can affect the protected circuit even when the unprotected circuit is unchanged.

The ZZ check is most compelling when the benefit in accepted-shot fidelity is clear and the acceptance remains high. That is the case for the contemporary model in the core results. In contrast, the 2016-inspired core case loses shots without a measurable fidelity gain, while the hyperbolic-inspired gain is small. The placement and double-check variants further show that choosing another ancilla or adding another check can make the circuit worse.

> **Recommendation.** Make ancilla and check placement a noise-aware compilation decision. The compiler should jointly consider connectivity, two-qubit gate count, circuit depth, and calibration-derived gate/readout error rather than choosing a route on gate count alone. Where possible, hardware layouts should provide low-error ancilla candidates close to commonly used port pairs.

This recommendation follows from the tested results, especially the contemporary model and the ancilla-placement variants. The evidence is limited to one port pair per finite model and simulated noise, so it supports a design direction rather than a claim that one architecture is universally superior.

10. Limitations

- All core results are simulated using backend-derived or synthetic noise models; they are not a controlled hardware comparison across three real processors.
- Only one fixed pair of output ports per model is evaluated. Other port pairs, circuit sizes, transpiler seeds, and calibration snapshots may produce different results.
- Native runs differ in gate set and noise assumptions. The common-noise experiment reduces this confound but still reflects one finite graph, one circuit family, and one fixed compilation seed.
- The benchmark noise models do not include every possible physical effect, such as all crosstalk and leakage mechanisms. The sensitivity sweep varies two-qubit error but is not a full noise-parameter study.
- Protected fidelity is conditional on passing the check. It must always be reported with acceptance and discarded shots.
- Some narrative text in the supplied notebook is inconsistent with its saved CSV tables. Before submission, restart the kernel, run all cells with fixed settings, regenerate every table and figure, and ensure the interpretation uses that final run. Keep the optional real-QPU cell separate and disable it during rehearsal unless a hardware run is intended.

11. Conclusion

The project implements the required nine-case Bell-state comparison and evaluates a ZZ parity check on three processor models. All nine core cases pass the challenge's fidelity-verification criterion. Under the saved native-noise results, the ZZ check improves the contemporary model from 0.9460 to 0.9704, but does not provide a meaningful gain on the 2016-inspired model and provides only a small gain on the hyperbolic-inspired model. The check adds three two-qubit gates, one SWAP, additional circuit depth, and discarded shots. Builder-level placement, common-noise, and sensitivity experiments show that useful error protection depends on the interplay of topology, noise, and compiler choices. The practical conclusion is to co-optimize ancilla placement, routing, and calibrated errors, while judging fidelity together with resource overhead and data yield.

References and Project Artifacts

1. **Qiskit Fall Fest 2026 Hackathon – Basic Track:** Supplied challenge brief defining the benchmark tasks, Bell-state verification criteria, required measurement settings, and submission requirements.
2. **QFF2026*Basic*Track_Solution.ipynb:** Supplied Jupyter Notebook containing the implementation, experimental procedures, analysis, and saved outputs.
3. **Supplied Output Files:** core_nine_cases.csv, task5_builder_variants.csv, task6_common_noise.csv, task6_native_vs_common.csv, task6_sensitivity_sweep.csv, counts.json, and settings.json. These files contain benchmark results, experimental data, measurement counts, and configuration settings.
4. **Artificial Intelligence Assistance:** AI tools were used as supplementary references for selected explanations, code assistance, and content refinement. The project implementation, experimental results, and conclusions were reviewed against the supplied challenge materials and project outputs.